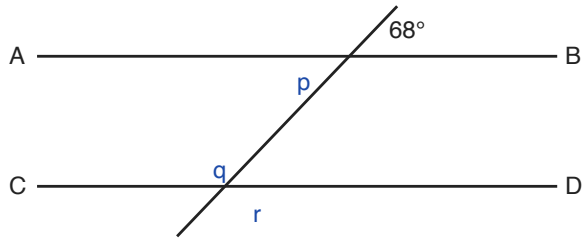
Shape and Space 1 — Angles · Triangles · Polygons · Similar triangles

Give a *reason* for every angle you find (e.g. "angles on a straight line", "alternate angles"). Diagrams are not drawn to scale. Marks in brackets. ★ = harder question.

A · ANGLES AND TRIANGLES

1 Lines AB and CD are parallel. A straight line crosses them as shown. Work out the angles p, q and r, giving a reason for each. [3]



$p = 68°$ — **vertically opposite** angles are equal.
$r = 68°$ — **corresponding angles** are equal ($AB \parallel CD$): r sits in the same position at the lower crossing as p does at the upper one.
$q = 180° - 68° = 112°$ — **angles on a straight line** (q and r) add up to $180°$; equally, p and q are **co-interior** angles, which add up to $180°$.

2 In triangle PQR, $PQ = PR$. Angle $Q = (2x + 5)°$ and angle $P = (x + 10)°$. Find x and the size of each angle. [3]

Isosceles: the base angles are equal, so angle R = angle $Q = (2x + 5)°$. Angle sum $180°$: $2(2x + 5) + x + 10 = 180 \Rightarrow 5x + 20 = 180 \Rightarrow x = 32$. Angles: $Q = R = 69°$, $P = 42°$ (sum 180 ✓).

3 In triangle ABC, the side BC is extended to a point D. Angle $ACD = 125°$ and angle $BAC = 70°$. Work out angle ABC, with a reason. [2]

The **exterior angle** of a triangle equals the sum of the two opposite interior angles: $125 = 70 + \angle ABC \Rightarrow \angle ABC = 55°$. (Or: $\angle ACB = 55°$ on a straight line, then $180 - 70 - 55 = 55°$.)

B · QUADRILATERALS AND POLYGONS

4 The angles of a quadrilateral are $3x°$, $(2x + 20)°$, $(x + 40)°$ and $90°$. Find x and the four angles. [3]

> Angles in a quadrilateral add to $360°$:
> $3x + 2x + 20 + x + 40 + 90 = 360 \Rightarrow 6x + 150 = 360 \Rightarrow x = 35$. Angles: $105°$, $90°$, $75°$, $90°$ (sum 360 ✓).

5 **a** Work out the interior angle of a regular decagon (10 sides). [4]

b A regular polygon has exterior angles of $24°$. How many sides does it have, and what is its interior angle?

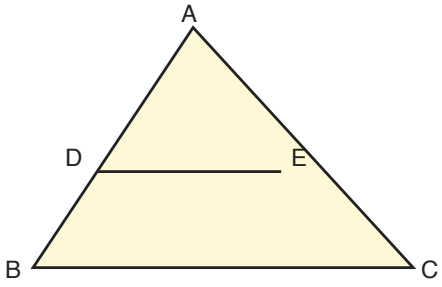
> **a** Exterior angle $= \dfrac{360}{10} = 36°$, so interior $= 180 - 36 = 144°$. (Or $\dfrac{(10-2) \times 180}{10} = 144°$.)
>
> **b** Exterior angles sum to $360°$: $n = \dfrac{360}{24} = 15$ sides. Interior $= 180 - 24 = 156°$.

6 A regular hexagon and a square share a side. At one end of the shared side, the hexagon and the square each have another side meeting at that vertex. Calculate the angle between those two sides, outside both shapes. Show your working. [3]

> Interior angle of a regular hexagon: $180 - \dfrac{360}{6} = 120°$. Interior angle of a square: $90°$.
> Angles around a point add to $360°$: $360 - 120 - 90 = 150°$.

C · SIMILAR TRIANGLES AND SCALE

7 In triangle ABC, D lies on AB and E lies on AC with DE parallel to BC. $AD = 6$ cm, $DB = 4$ cm, $DE = 9$ cm and $AE = 7.5$ cm. [4]



a Explain why triangles ADE and ABC are similar.

b Work out the length BC.

c Work out the length EC.

> **a** Angle A is common; $\angle ADE = \angle ABC$ and $\angle AED = \angle ACB$ (corresponding angles, $DE \parallel BC$). Three equal angles → similar.
>
> **b** Scale factor from ADE to ABC: $\dfrac{AB}{AD} = \dfrac{10}{6} = \dfrac{5}{3}$. So $BC = 9 \times \dfrac{5}{3} = 15$ cm.
>
> **c** $AC = 7.5 \times \dfrac{5}{3} = 12.5$ cm, so $EC = 12.5 - 7.5 = 5$ cm.

8 A map has scale $1 : 50\,000$. [3]

a Two villages are 7.4 cm apart on the map. How far apart are they in real life, in km?

b A road is 12 km long. How long is it on the map?

> **a** $7.4 \times 50\,000 = 370\,000$ cm $= 3\,700$ m $= 3.7$ km.
> **b** 12 km $= 1\,200\,000$ cm; $\div 50\,000 = 24$ cm.

9 Describe, step by step, how to construct with compasses and a ruler [2]

a the perpendicular bisector of a line segment AB of length 6 cm;

b an angle of $60°$.

> **a** Open the compasses to more than half of AB. With the point on A, draw arcs above and below the line; keep the same radius, point on B, draw two more arcs crossing the first pair. Join the two crossing points — this line cuts AB at right angles through its midpoint.
> **b** Draw a line and mark a point O on it. With the point on O draw an arc crossing the line at X. Keep the radius, put the point on X and draw an arc crossing the first arc at Y. Join OY: angle $XOY = 60°$ (triangle OXY is equilateral).

10 At the same moment, a 1.8 m tall person casts a shadow 2.4 m long and a tree casts a shadow 14 m long. Use similar triangles to find the height of the tree. [3]

> The sun's rays make the same angle with the ground for both, so the two right-angled triangles (height, shadow) are similar. $\dfrac{\text{height}}{\text{shadow}}$ is the same:
> $\dfrac{h}{14} = \dfrac{1.8}{2.4} \Rightarrow h = 14 \times 0.75 = 10.5$ m.

Total: 30 marks

Reasons to quote: angles on a line 180° · around a point 360° · vertically opposite · alternate / corresponding / co-interior (parallel lines) · triangle 180° · quadrilateral 360° · exterior angles of a polygon 360°.